Bibliographic search

a.k.a. Fantastic publications and where to find them

INFO B302 – Introduction à la démarche scientifique (IDS)

Xavier Devroey <xavier.devroey@unamur.be>

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State-of-the-art: why does it matter?

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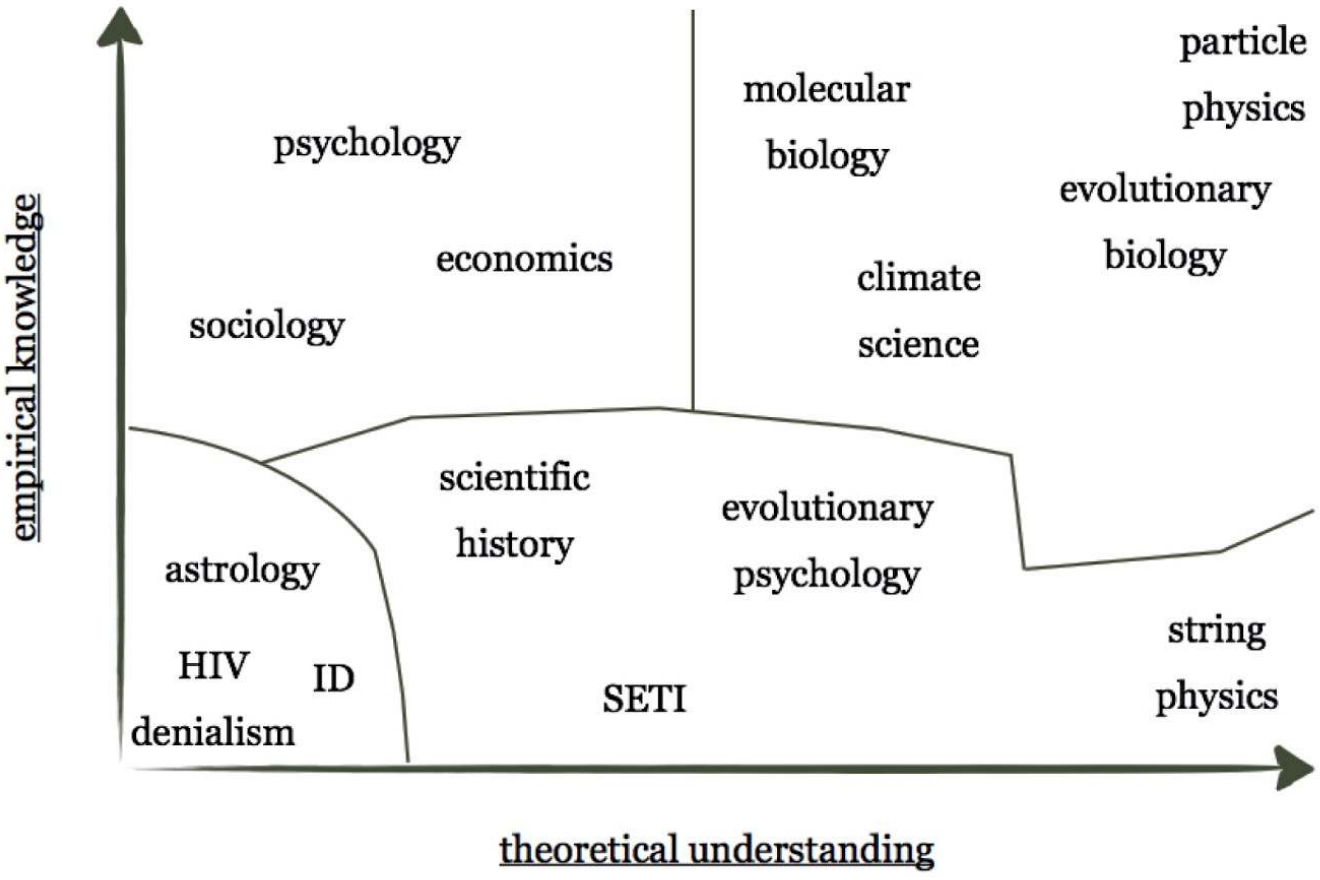
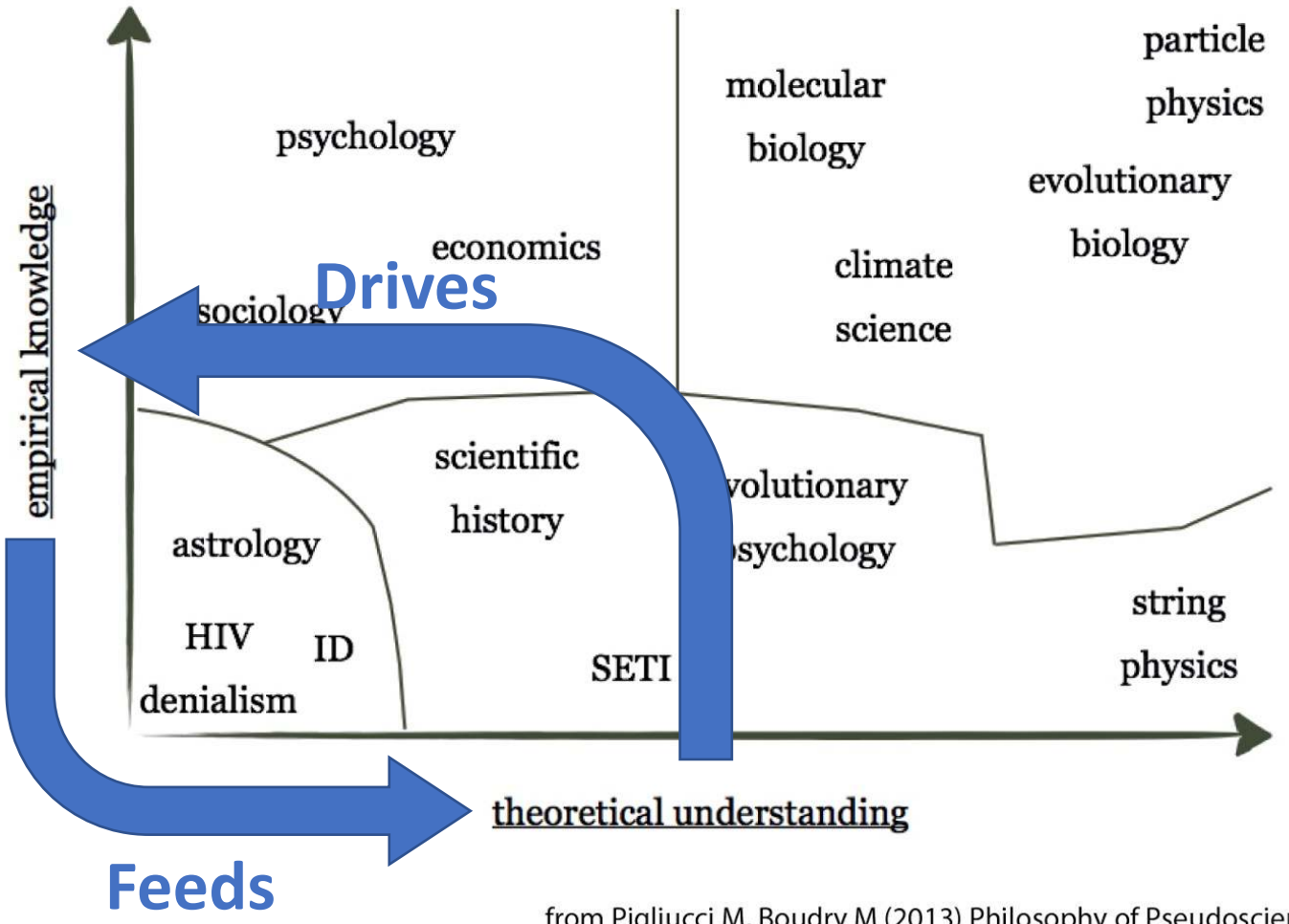
The demarcation problem



from Pigliucci M, Boudry M (2013) Philosophy of Pseudoscience: Reconsidering the Demarcation Problem; University of Chicago Press. with permission of M Pigliucci.

The demarcation problem



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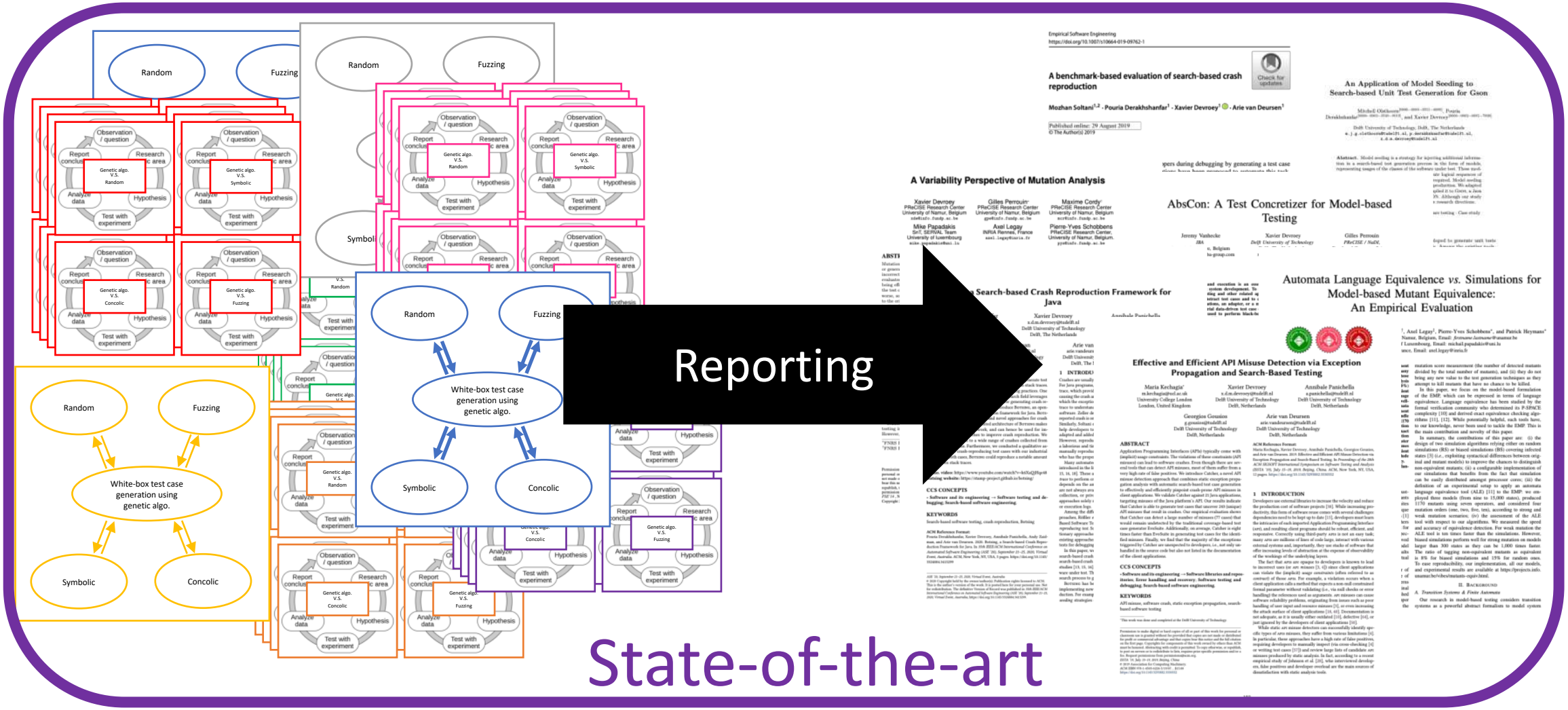
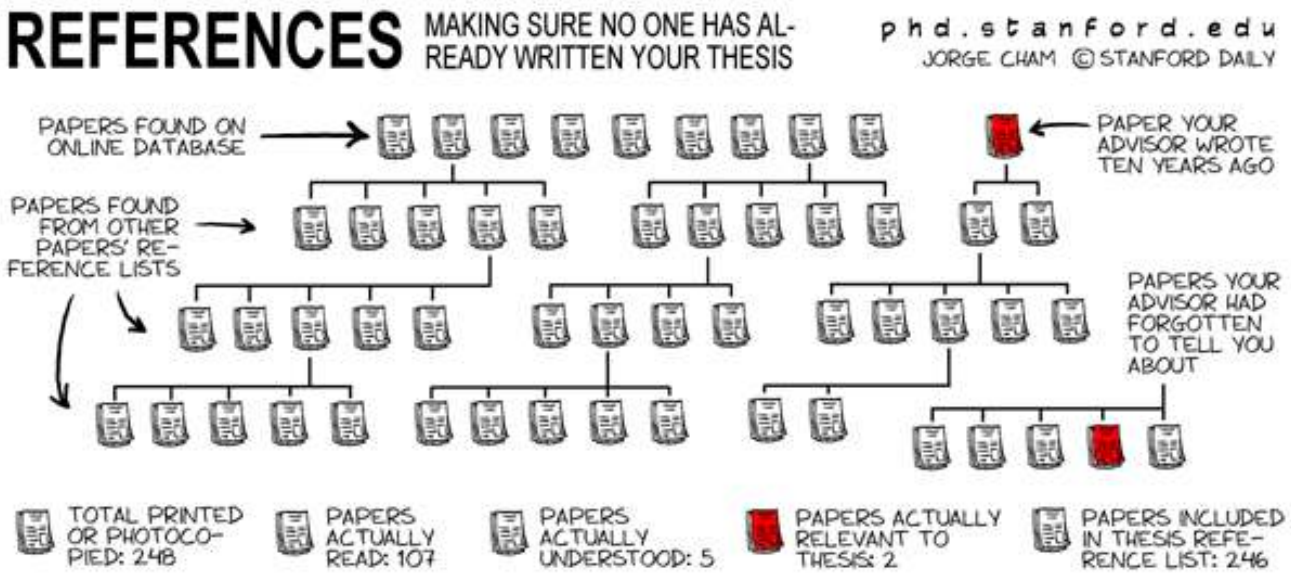
Reporting in (peer-reviewed) publications



Citing sources



… but how does one find those references?

How to find publications?

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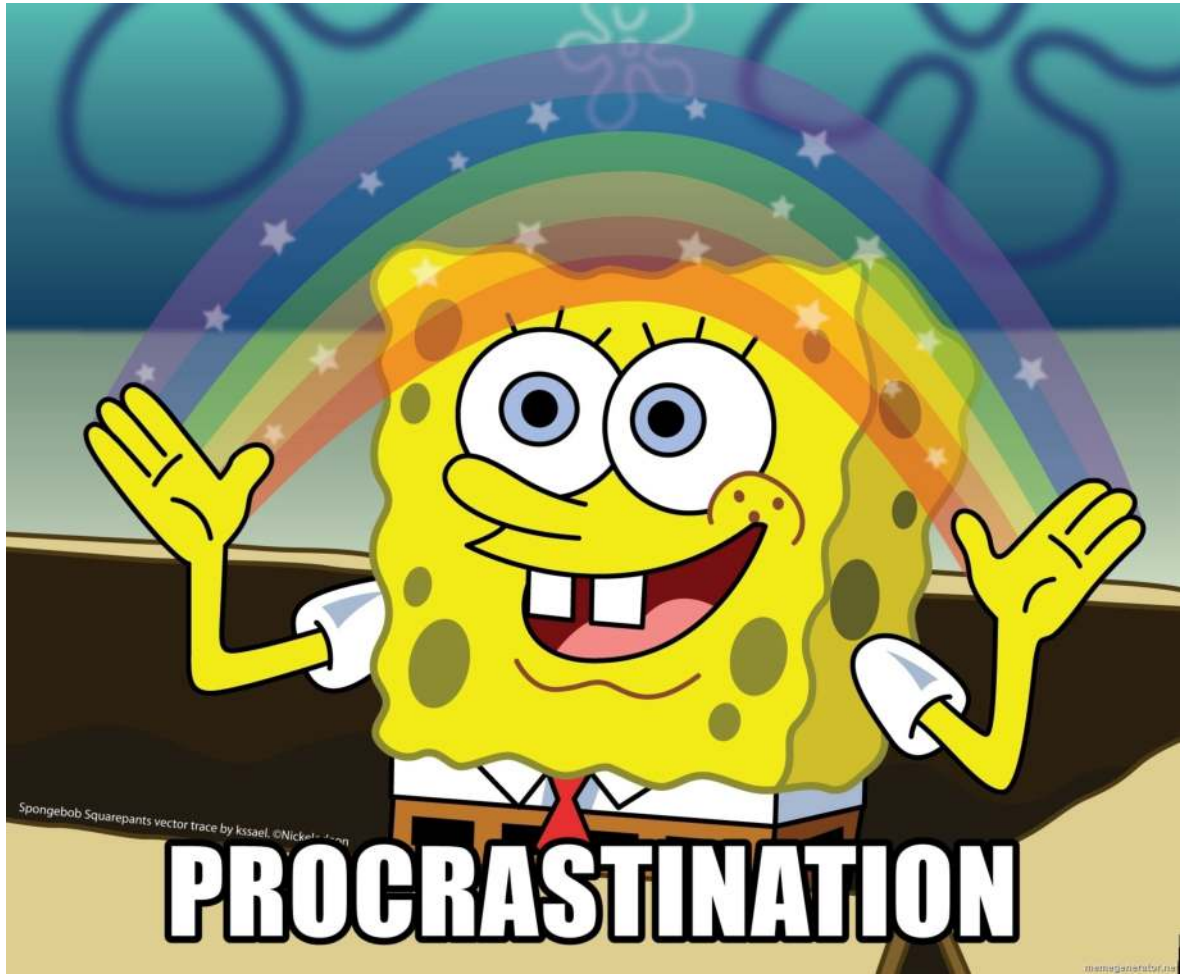
Try finding information about

Chi, Michelene TH, and Randi Daimon Koeske. "Network representation of a child's dinosaur knowledge." Developmental psychology 19.1 (1983): 29.

Scenario 1

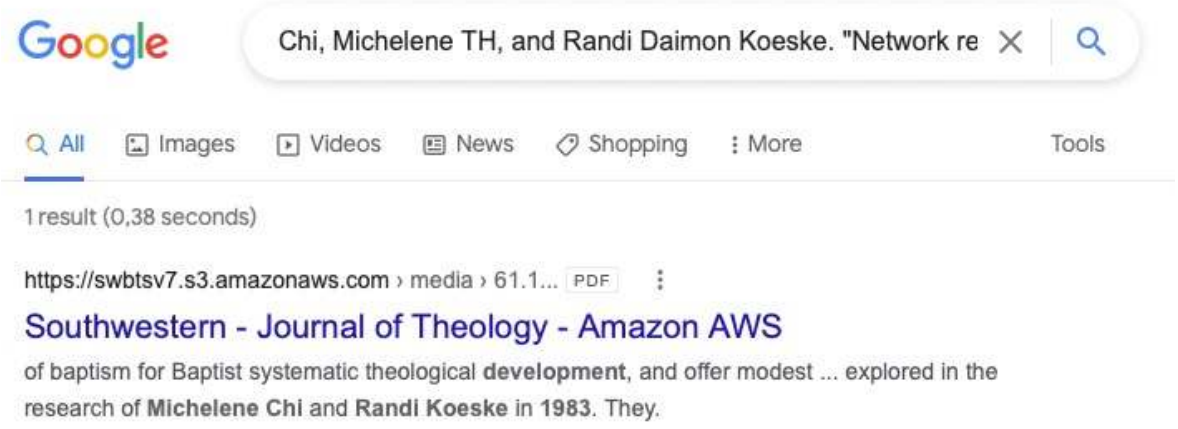
- You did not do the activity because <insert poor excuse here>

- … Remember that participation is continuously graded!



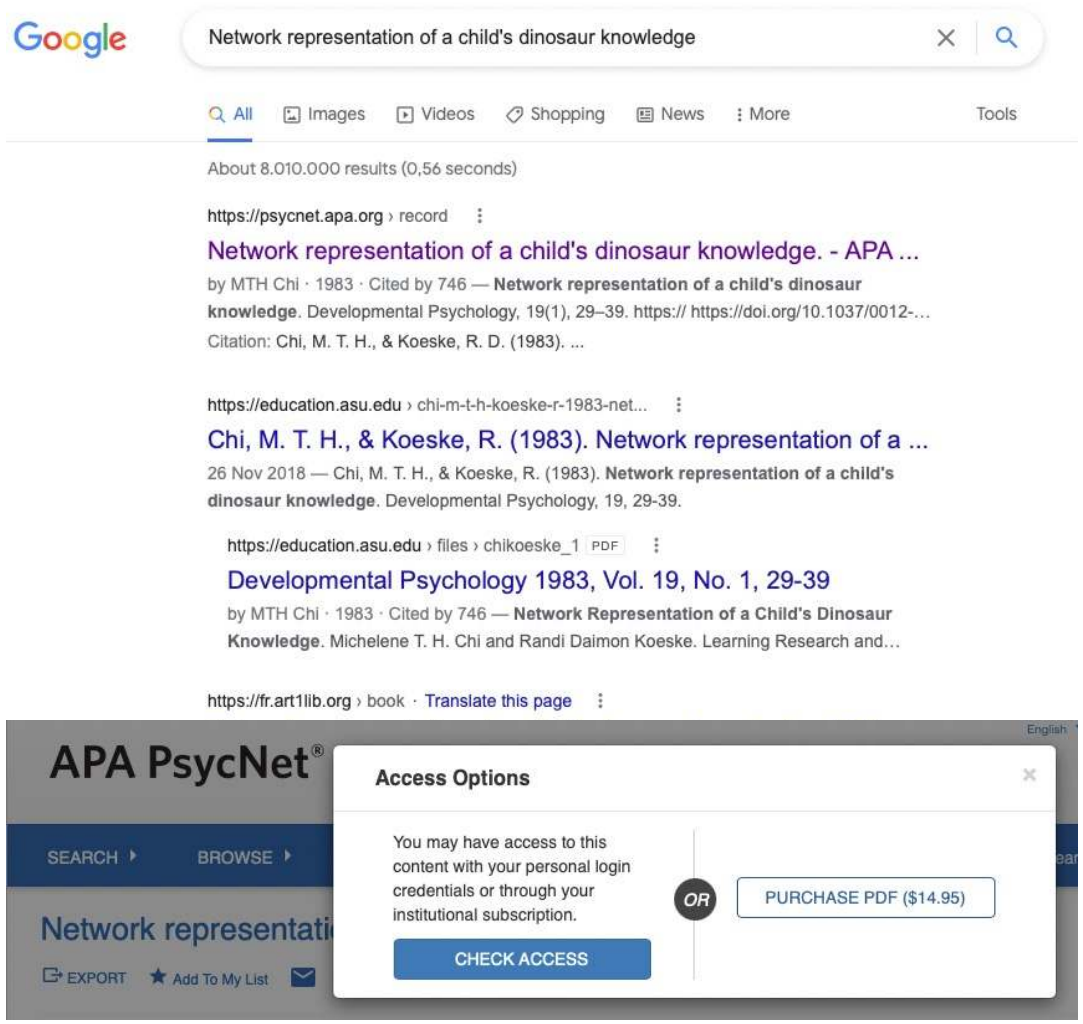
Scenario 2

- You pasted the whole bibliographic reference in Google
- Congrats! You got one results. A 129-page PDF about theology.



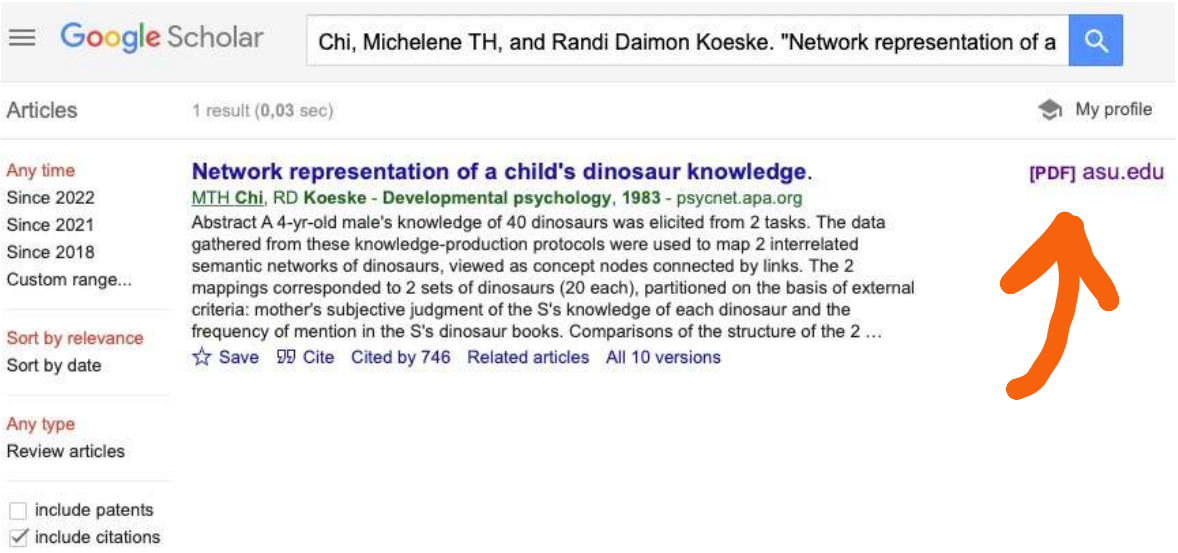
Scenario 3

- You searched the article title in Google
- Congrats! You get several results about the right article
- … and you can get the article for $14.95!



Scenario 4

- You searched either the full reference or the title on Google Scholar (https://scholar.google.com)
- Congrats! You found one single entry with the right article!
- … plus, you get the article PDF for free



How to find publications?

(the systematic way)

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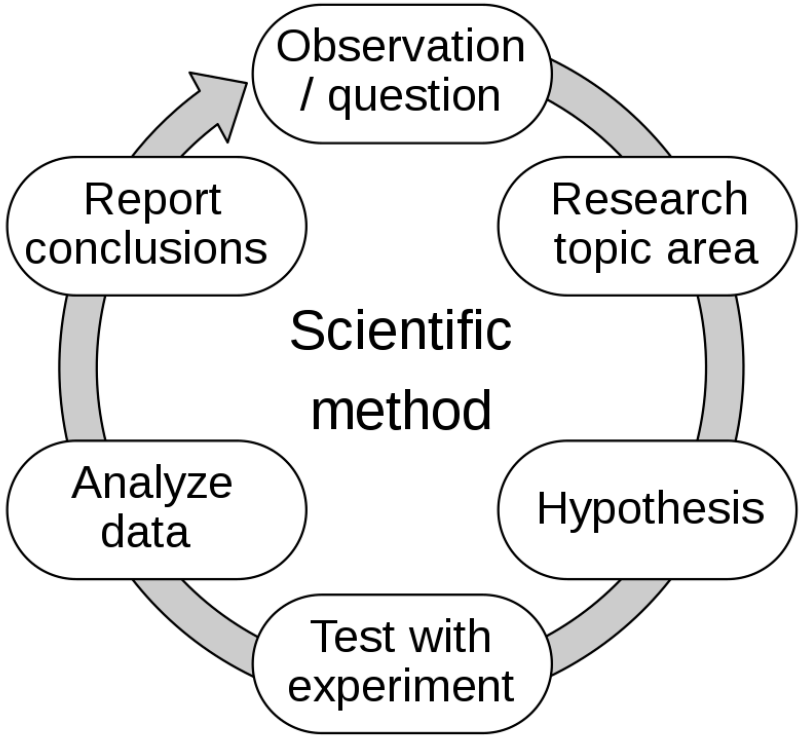
What is the current state-of-the-art of automated test case generation

Report the findings, including the (full) process followed to conduct the study



White-box test case generation

Data synthesis from the selected studies to collate and summarise the results qualitatively or quantitatively

- Most literature concerns search-based approaches (?)
- There are few case studies on the deployment of automated test case generation in industry
- …

- Do a systematic search of the literature and identify relevant studies.
- Assess the quality of the identified studies

Systematic Literature Review (SLR)

- To summarise the existing evidence concerning a treatment or technology
  - For instance to summarise the empirical evidence of the benefits and limitations of a specific agile method
- To identify any gaps in current research
- To provide a framework/background in order to appropriately position new research activities
- To examine the extent to which empirical evidence supports/contradicts theoretical hypotheses and assist the generation of new hypotheses

Guidelines for performing Systematic Literature Reviews in Software Engineering

Version 2.3

EBSE Technical Report
EBSE-2007-01

Software Engineering Group
School of Computer Science and Mathematics
Keele University
Keele, Staffs
ST5 5BG, UK

and

Department of Computer Science
University of Durham
Durham,
UK

9 July, 2007

Systematic Literature Review

1. Plan the review

1. Specify research questions
2. Develop review protocol
3. Validate review protocol

2. Conduct the review

1. Identify relevant research
2. Select primary studie
3. Assess study quality
4. Extract required data
5. Synthesise data

3. Document the review

1. Write review report
2. Validate report

Advantages SLRs

- The well-defined methodology makes it less likely that the results of the literature are biased
  - Although it does not protect against publication bias in the primary studies
- They can provide information about the effects of some phenomenon across a wide range of settings and empirical methods
  - If studies give consistent results, systematic reviews provide evidence that the phenomenon is robust and transferable
  - If the studies give inconsistent results, sources of variation can be studied

Disadvantage of SLRs

- It requires considerably more effort than traditional literature reviews

Identifying relevant literature
(the pragmatic way)

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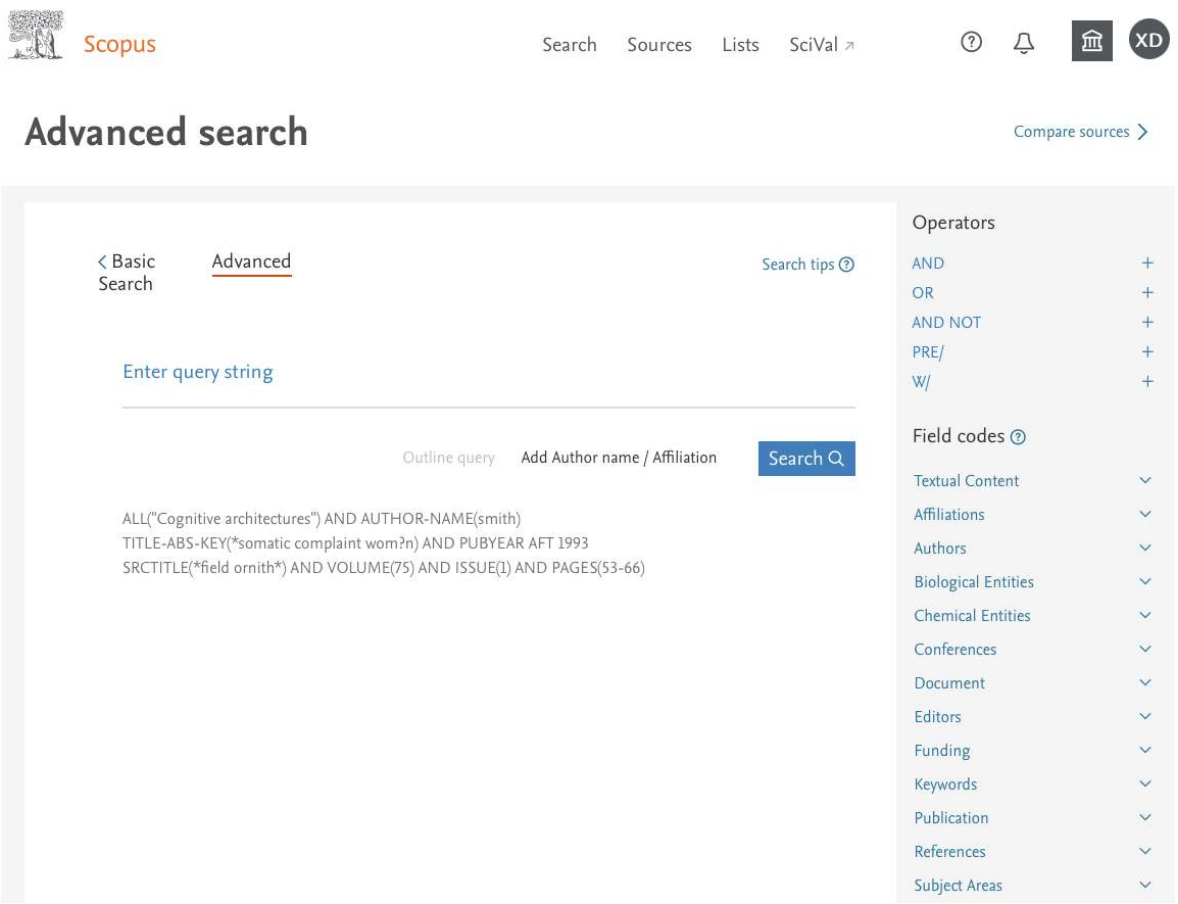
Literature search engines

- General purpose
  - Google Scholar (https://scholar.google.com)
  - Microsoft Academic (https://academic.microsoft.com)
  - Scopus (https://www.scopus.com/)
- Publishers
  - ACM Digital Library (https://dl.acm.org)
  - IEEEXplore (https://ieeexplore.ieee.org/)
  - ScienceDirect (https://www.sciencedirect.com)
  - Springer Link (https://link.springer.com)

Search string

- Keywords
  - (test OR testing) AND (“search-based” OR “evolutionary computation”)
- In
  - Title, abstract, keywords, full paper, …
- Authors
- Publication year
- …



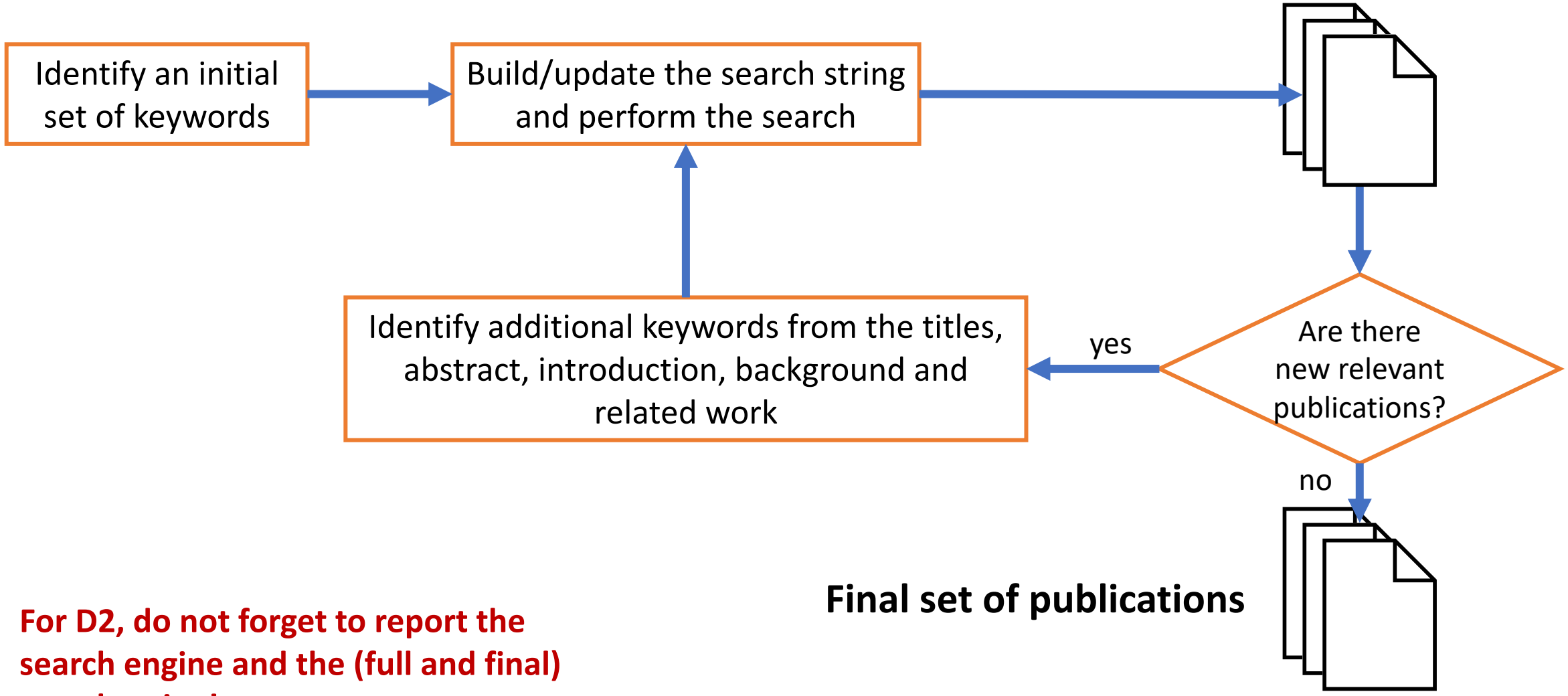
Identifying relevant keywords

- From a general description (…like the ones you received 😉)
- From an informal list of papers
  - Provided by other researchers
  - Collected from a conference or workshop in the topic
- From a formal list of papers
  - Systematic literature review
  - Systematic mapping study
  - …
  - (those also describe the search strings used for the study)

The PICOC framework

- Population: who/what is the target affected by the intervention?
  - Computer science students with little to no training on literature reviews
- Intervention: what is done on the population?
  - Course on bibliographic search
- Comparison: what are the alternatives?
  - None in this case
- Outcome: what is the result of the intervention?
  - Improved understanding of how to conduct a literature review
- Context: in what context does the intervention take place?
  - Computer science program

Building the search string



For D2, do not forget to report the search engine and the (full and final) search string!

Inclusion and exclusion

- Goal: identification of relevant publications
- Criteria based on the content
  - Relevant to the research domain considered
  - Describing elements the background
    - Any element necessary to understand what the research is about
  - Describing relevant related work
    - Similar approach for a different problem
    - Same problem using a different approach
    - Same problem using the same approach but with a different goal
    - Same problem using the same approach with the same goal

Inclusion and exclusion

- Criteria based on meta-data
  - Peer-reviewed
  - Grey literature (non-peer reviewed)
  - Type of publication
    - Books
    - Research articles
      - Short paper
      - Full paper
    - Master/PhD thesis
    - Technical report
    - …
  - Venue
    - Conferences
    - Journals

Application of inclusion/exclusion criteria

1. Read the title, abstract, keywords
2. Decide if the paper is relevant or not
3. In case of doubt, read the introduction and conclusion to make a decision
4. (Track your decision, for instance, in a spreadsheet)

For D2, include peer-reviewed research articles only and track and report your decisions.

Snowballing

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Citations

- Goal: show that you took words, ideas, figures, images, etc. from another place.
- Citations are a short way to uniquely identify a published
  - Include a DOI whenever possible
- Located at the end of the document or in footnotes

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Based on https://libguides.mit.edu/citing

Snowballing

- Technique used to identify additional publications
- Looking at citations graph
  - Cited by the paper (prior work)
    - References list
  - Citing the paper (derivative work)
    - Via a literature search engine

Search-based crash reproduction using behavioural model seeding
P Derakhshanfar, X Devroey, G Perrouin… - Software Testing …, 2020 - Wiley Online Library
Search-based crash reproduction approaches assist developers during debugging by generating a test case, which reproduces a crash given its stack trace. One of the fundamental steps of this approach is creating objects needed to trigger the crash. One way to overcome this limitation is seeding: using information about the application during the search process. With seeding the existing usages of classes can be used in the search process to produce realistic sequences of method calls, which create the required objects. In …
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Search-based crash reproduction using behavioural model seeding
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Approaches for automatic crash reproduction aim to generate test cases that reproduce crashes starting from the crash stack traces. These tests help developers during their debugging practices. One of the most promising techniques in this research field leverages …
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Commonality-Driven Unit Test Generation
B Evers, P Derakhshanfar, X Devroey… - … Symposium on Search …, 2020 - Springer
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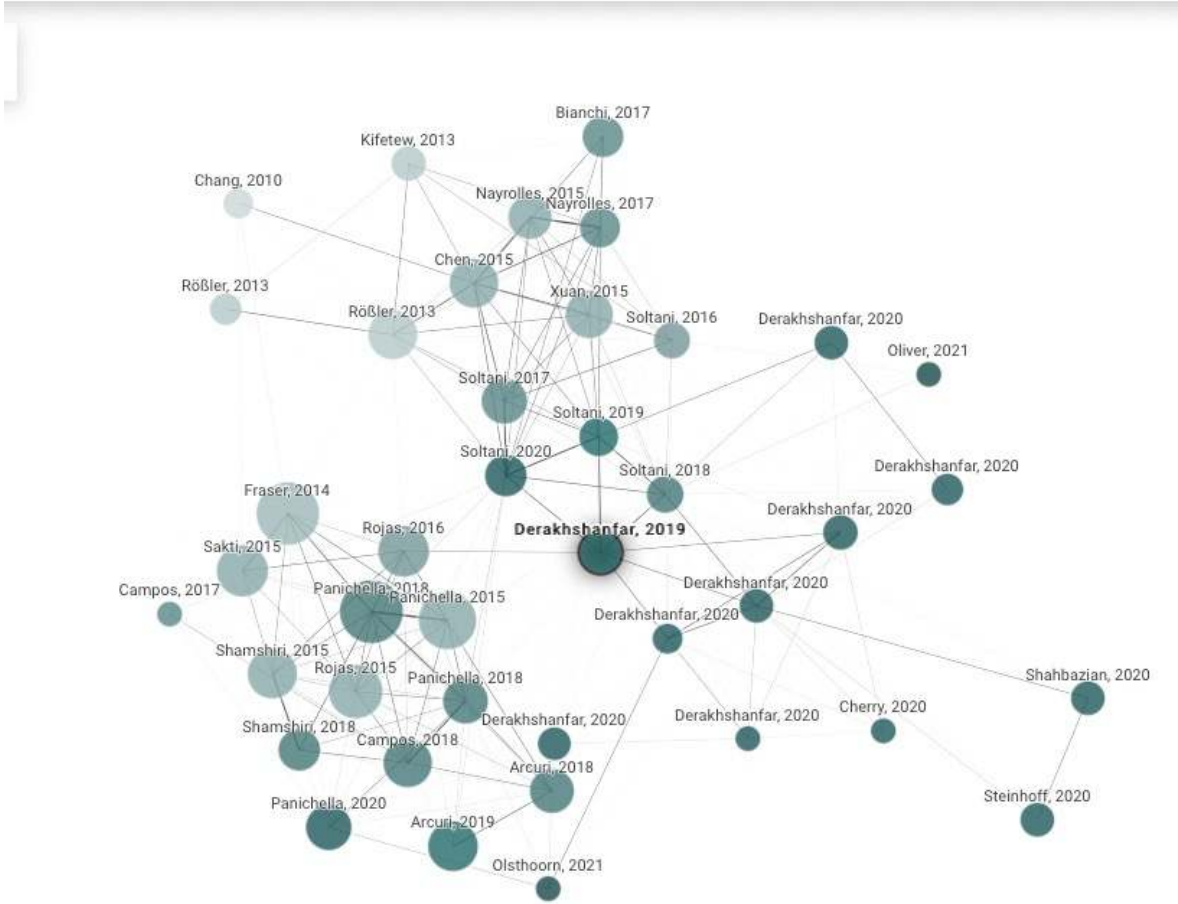
Citation graphs

- Used to identify research communities
- Built from data collected in literature search engines
- Useful to quickly identify important seminal works

Search-based crash reproduction using behavioural model seeding



https://www.connectedpapers.com/

Gotta find’em all?

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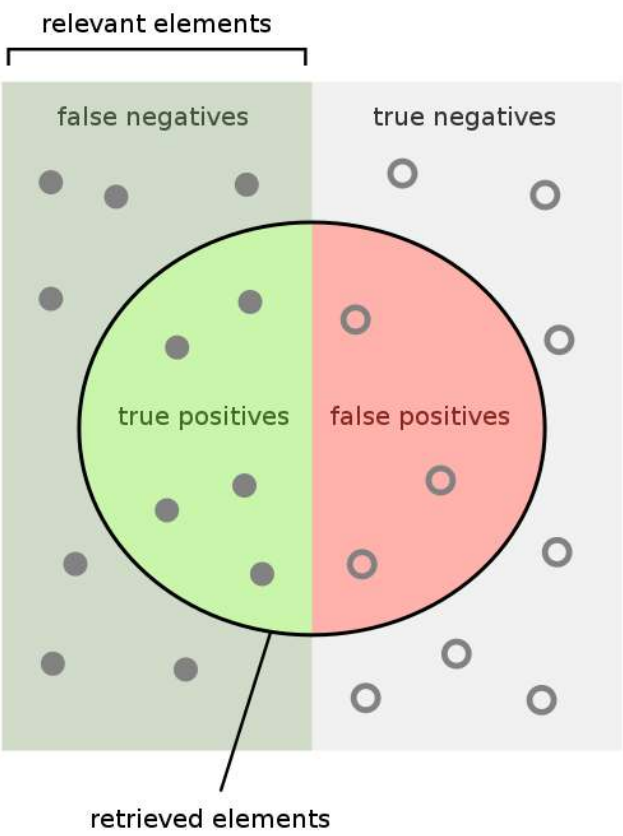
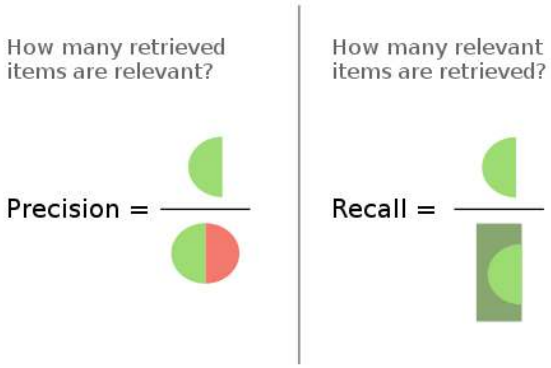
Precision and Recall

- **Precision** measures the number of relevant articles among those returned by the search strategy
- **Recall** measures the number of relevant articles found among all relevant articles that exist
- Relevant elements is unknown (using an approximation)



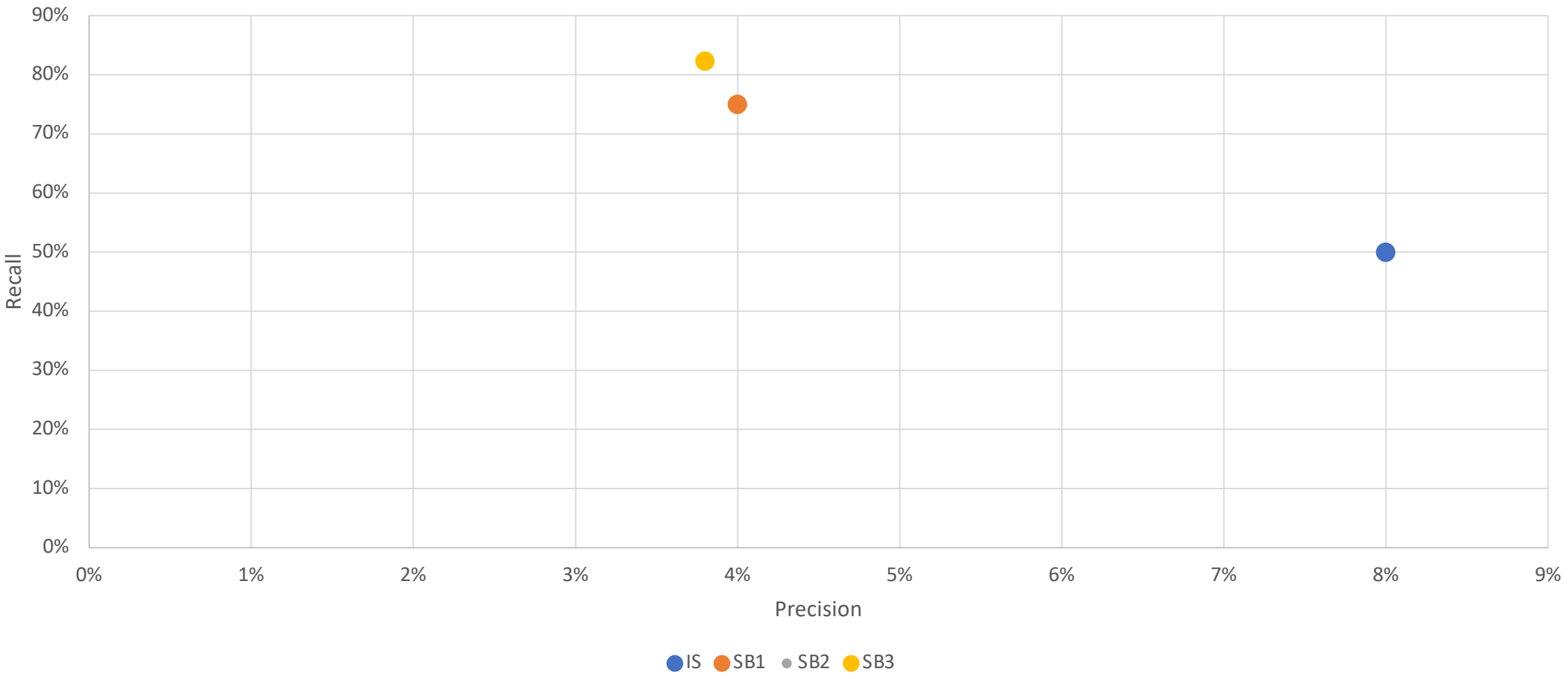


Example

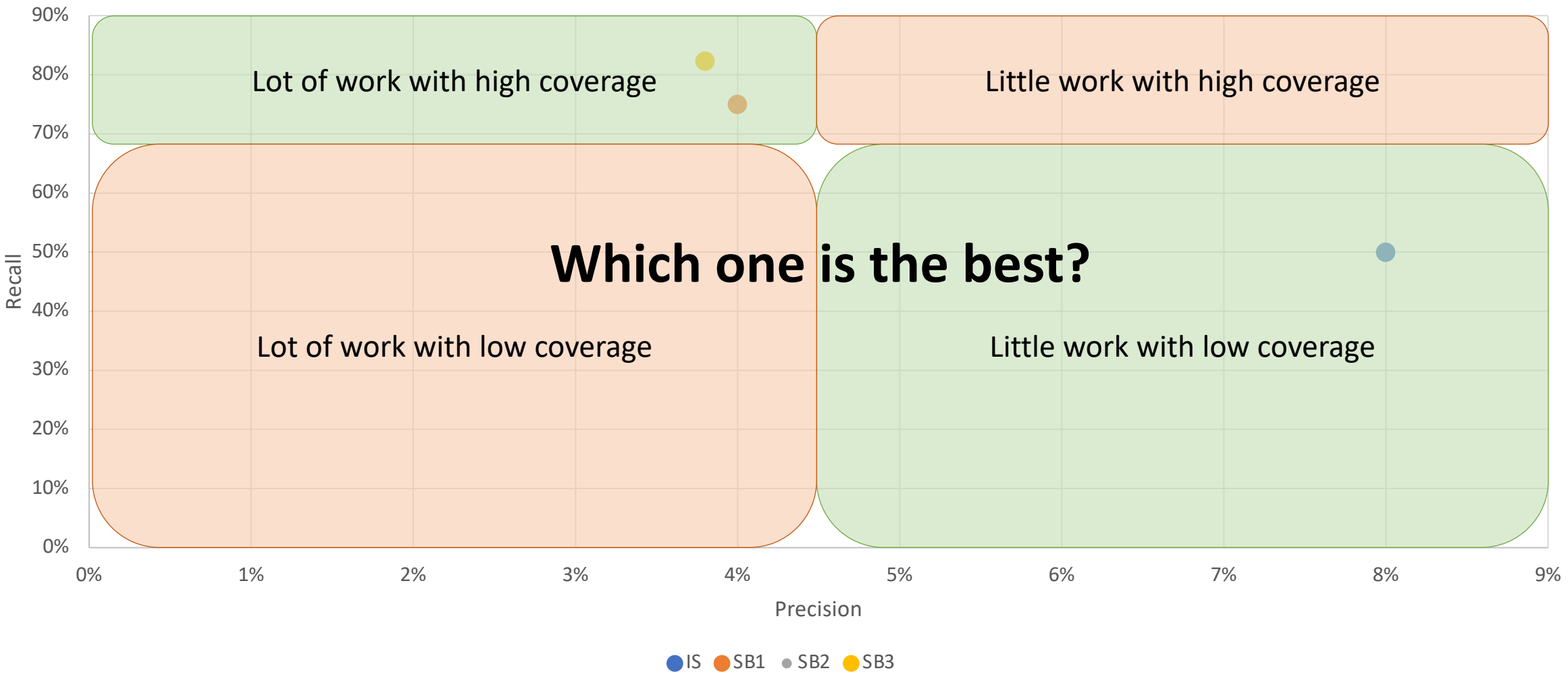
Strategy	Results	Relevant	Precision*	Recall*
Initial search	250	20	8%	50%
+ snowballing	250 + 500	20 + 10	4%	75%
+ snowballing	750 + 100	30 + 3	3.8%	82.4%
+ snowballing	850 + 20	33 + 0	3.8%	82.3%

*Assuming a total number of 40 relevant papers

Example


90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Recall
0%
1%
2%
3%
4%
5%
6%
7%
8%
9%
Precision
IS
SB1
SB2
SB3

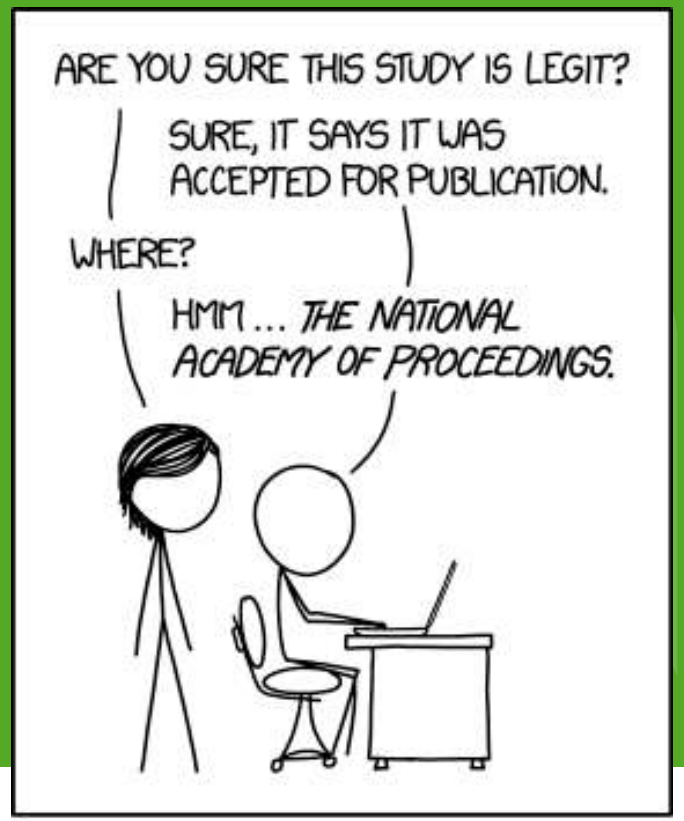
Example


90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Recall
0%
1%
2%
3%
4%
5%
6%
7%
8%
9%
Precision
Lot of work with high coverage
Little work with high coverage
Which one is the best?
Lot of work with low coverage
Little work with low coverage
IS
SB1
SB2
SB3

Precision and Recall

- High precision and recall is not possible
- Background and related work
  - Requires a representative set of publications (mainly research articles and books)
  - Focus on most cited references first
- Literature review
  - Aim for the largest recall possible
- Systematic literature review
  - Aim for the largest recall possible and discuss strategies to do so

Pitfalls to avoid



IF SOMETHING IS FORMATTED LIKE A SERIOUS SCIENTIFIC PAPER, IT CAN TAKE ME A WHILE TO REALIZE IT ISN'T ONE.

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Predatory journals and publishers

American Journal of
Biomedical Science & Research

www.biomedgrid.com
ISSN: 2642-1747

Mini Review

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Cyllage City COVID-19 Outbreak Linked to Zubat Consumption

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To Cite This Article: *Mattan Schlomi, Cyllage City COVID-19 Outbreak Linked to Zubat Consumption. 2020 - 8(2). AJBSR.MS.ID.001256. DOI: 10.34297/AJBSR.2020.08.001256.*

Received: March 14, 2020; **Published:** March 18, 2020

Abstract

A localized outbreak of COVID-19 pneumonia in Cyllage City among a population with no history of travel to infected regions has been tentatively linked to consumption of the regional insectivorous zubat (*Vespertilio caeruleus sineoculus*, Pseudochiroptera), supported by amino acid sequence similarity between SARS-CoV-2 and a related coronavirus from the bats. An increased need for surveillance of bat viruses in Kalos is warranted.

Review

Coronavirus disease 2019 (Covid-19) is caused by a virus of the Coronavirus family, SARS-CoV-2 [1]. The outbreak began in the city of Wuhan, China, and has since spread to every inhabited continent, with over 100000 cases and a climbing death toll [2]. Identifying the origin of the virus has proven difficult, in part due to misinformation associated with video of a Chinese travel blogger eating bat soup in Palau falsely attributed to the seafood market in Wuhan at the epicenter of the outbreak [3]. Molecular evidence has tentatively traced the origins of the virus to wild vertebrates including snakes [4-7], pangolins [8-11], pikas [12], sentrets [13-15], and kappas [16]. The former two are the most likely animal consumed in Wuhan that sparked the zoonotic event in Patient Zero [2,7-8].

However, most comparative virological evidence to date places SARS-CoV-2's origin within the bats: this evidence includes analysis of the RNA sequence of the virus' genetic material [17-20], amino acid sequencing of the capsid proteins [21-24], and fatty acid analysis of the lipid envelope [25]. This finding is similar to that of SARS-CoV-1, which was linked to consumption of a masked palm civet cat in Guangdong province but whose viral RNA is most closely related to that of horseshoe bats (genus Rhinolophus) [26,27]. Both bats and civets were in close contact with each other in the market where the SARS outbreak began and could have transmitted the virus amongst each other there, but another possibility is that the civets had been infected with the bat virus beforehand [28]. Bats are known to harbor many viruses and are the origin of several zoonotic diseases, such as Nipah virus and Ebola [26,29-30], so virologists are not surprised that SARS-CoV-2 appears to have originated in bats.

Most outbreaks of COVID-19 outside China have been traced to travellers from Wuhan or those who came in contact with them [31-33], and community spreading of the virus has been confirmed [34,35]. To this literature we add a report from Cyllage City in the Kalos region, France, where an outbreak of the densely populated metropolis has to date produced 420 confirmed infections with seven deaths, all in the elderly and those with pre-existing conditions [36-38], matching the widely reported 2% fatality estimates for COVID-19 [2]. The outbreak is remarkable as no residents are known to have travelled to infected areas in recent months; how-

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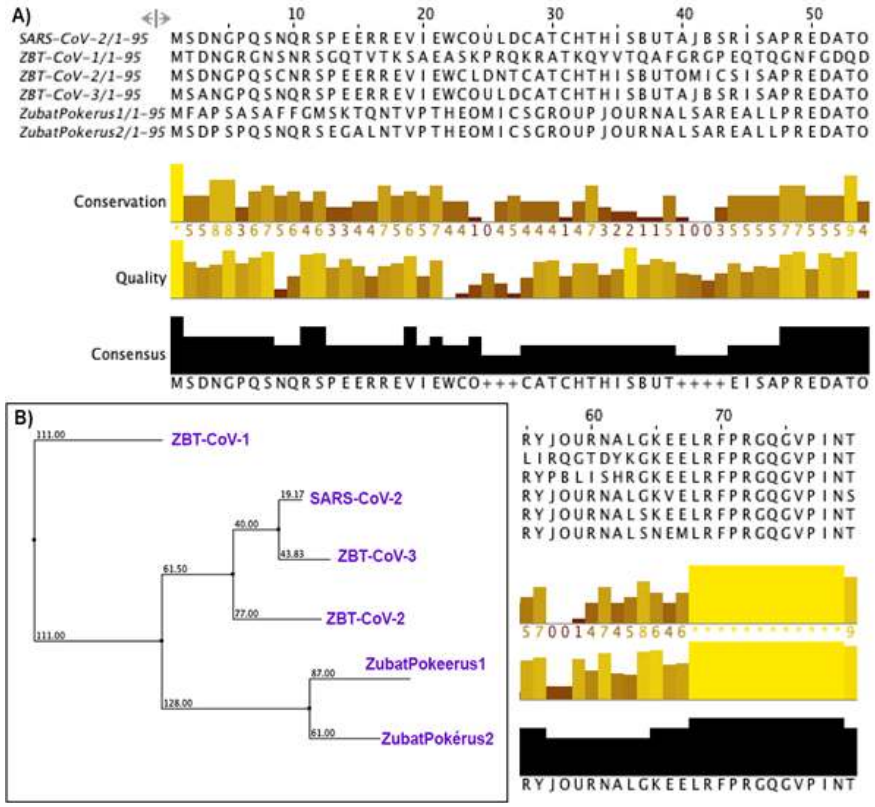

Figure 1: Nucleoprotein Sequence Comparisons of SARS-CoV-2 and Zubat Viruses

http://doi.org/10.34297/AJBSR.2020.08.001256

Low quality publication venues

- Predatory journals and predatory publishers
  - Charge publication fees to the authors without checking quality
- Fake conferences
- Solution: check that the journal is not listed as predatory or that the editorial board/conference organisers are well-known
  - https://beallslist.net

Literature search engine

- Google Scholar
  - Indexes everything
  - … including grey (non-peer-reviewed) literature and irrelevant papers
- Scopus
  - Cleaner database
  - Not always up-to-date with the latest publications
- Solutions
  - Combine multiple search engines
  - Confirm publications using https://dblp.uni-trier.de

Wrong keywords

- Different research communities address the same topic but with different keywords
  - For instance, *feature* in *software product line engineering* and *configuration option* in *software architecture*
- Different research communities address different topics with the same keyword
  - For instance, *feature* in *requirements engineering* and *machine learning*
- Solution: ask researchers of different domains or select papers from different research communities.

Problems with the search strategy

- Wrong scope size
  - Solution: adjust the scope after reading a few research papers on the topic of interest
- Inclusion and exclusion criteria are too vague
  - Solution: formulate binary criteria
  - Solution: do a cross check by asking another researcher to apply the criteria
- Do not know where to start with the keywords selection
  - Solution: ask a senior researcher in the field to point you a few relevant publications

D2 Bibliographie

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Contenu

- **Une liste d’articles**
  - 30 articles de recherche publiés et ayant été évalués par les pairs
  - Pas de preprints (venant d’ArXiv)
  - En cas de doute, posez-nous la question
- Information et template en LaTeX à utiliser disponibles sur Webcampus